Capstone Project Report

Autonomous Driving: Object Detection and Tesla Autopilot Data Analysis

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Abstract

This project addresses the growing significance of Artificial Intelligence in the domain of autonomous driving. It consists of two parts: (1) building a Convolutional Neural Network (CNN)-based object detection model to classify and localize vehicles in road images, and (2) analyzing Tesla Autopilot accident datasets to evaluate the impact of autonomous features on road safety. The results demonstrate the ability of AI to assist in real-time vehicle detection and provide insights into accident trends, thus contributing to the development of safer intelligent transportation systems.

Introduction

Autonomous vehicles (AV) and Intelligent Transport Systems (ITS) are revolutionizing road transport. Automatic detection of vehicles in real time is crucial for enabling safe navigation, traffic monitoring, and incident response. Companies such as Tesla are pioneers in integrating AI into self-driving technology, but the safety of such systems remains under scrutiny. This project combines deep learning-based computer vision and data analysis to explore both technological feasibility and real-world safety implications.

Problem Statement

The project is divided into two parts:

Part 1: Develop a CNN-based deep learning model that detects vehicles in images and localizes them with bounding boxes.

Part 2: Analyze Tesla Autopilot accident datasets to assess the role of autonomous driving features in road safety.

Dataset Description

Part 1 Dataset: Images.zip, a collection of images containing vehicles for training object detection models.
Part 2 Dataset: Tesla-Deaths.csv, containing accident-related information such as

accident year, location, number of deaths, Tesla model, involvement of cyclists/pedestrians, and whether the event was autopilot verified.

Methodology

Part 1: CNN Object Detection

1. Preprocessing: The dataset was organized into train and test folders. Images were resized and augmented to improve generalization.
2. Model Design: A CNN-based architecture was implemented using TensorFlow/Keras. Layers included convolution, pooling, and fully connected layers.
3. Training: The model was trained over multiple epochs with a decreasing loss trend, indicating convergence.
4. Evaluation: Accuracy and loss metrics were monitored. The model successfully detected vehicles in unseen images.
5. Inference: Bounding boxes were drawn on vehicles, validating the model's object detection capability.

Part 2: Tesla Accident Data Analysis

1. Data Cleaning: Missing values, duplicates, and irrelevant columns were handled.
2. Exploratory Data Analysis: Accident trends were visualized by year, state, and Tesla model.
3. Key Questions: The analysis focused on number of victims per accident, driver fatalities, autopilot-related deaths, and involvement of cyclists/pedestrians.
4. Correlation Analysis: Relationships among numeric variables (year, deaths, occupants) were studied using a heatmap.

Results & Discussion

Part 1 Results

The CNN model demonstrated effective object detection capabilities. Training logs showed improvement in accuracy and reduction in loss. Sample inference outputs displayed bounding boxes correctly identifying multiple vehicles within the same frame. This highlights the potential of CNNs in real-time autonomous driving systems.

Part 2 Results

1. Accident trends increased significantly after 2016, with a peak around 2019-2020.
2. California and Florida recorded the highest number of Tesla-related incidents.
3. Model S had the highest involvement in fatal accidents compared to other Tesla models.
4. Verified Tesla Autopilot deaths were identified, raising concerns about the safety of autonomous driving.

5. Correlation analysis showed moderate associations between event year and number of deaths.

Conclusion & Future Enhancement Work

This project successfully implemented an AI-based vehicle detection system and conducted a data-driven analysis of Tesla Autopilot safety. The CNN model proved effective for real-time object detection tasks, while the accident dataset analysis provided valuable insights into autonomous vehicle safety concerns. Future work may involve:
- Deploying real-time detection on embedded devices for AVs.
- Using more advanced architectures such as YOLOv5/YOLOv8.
- Conducting deeper statistical analysis on Tesla accident data with larger datasets.
- Exploring causal inference to establish links between autopilot features and accident severity.

References

1. Dataset: Tesla-Deaths.csv (public dataset)
2. TensorFlow and Keras Documentation
3. Research papers on CNN-based Object Detection
4. Tesla Autopilot Safety Reports (NHTSA)